ML-ESS

Multi-Agent AI Research Assistant

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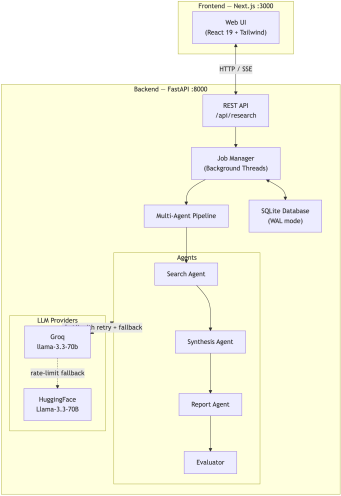
The bottleneck

- Manual research: hours of querying, reading, drafting
- No reproducibility or quality scores

ML-ESS automates it

- One question in
- Cited, scored report out
- Web UI ▪ REST API ▪ Telegram

FastAPI backend on Render; Next.js frontend on Vercel. SQLite WAL for job persistence.



Search Agent

Generates queries, scrapes Tavily/DDG, extracts evidence triples.

Synthesis Agent

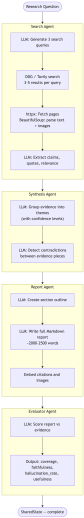
Groups evidence into themes; detects contradictions.

Report Agent

Outline-first, then ~2 000-word cited Markdown report.

Evaluator

Scores Coverage, Faithfulness, Hallucination, Usefulness.



Question: *"What are the main trade-offs between CNNs and Vision Transformers for medical imaging?"*

Search

3 queries ▪ 14 sources

47 evidence fragments extracted

Synthesis

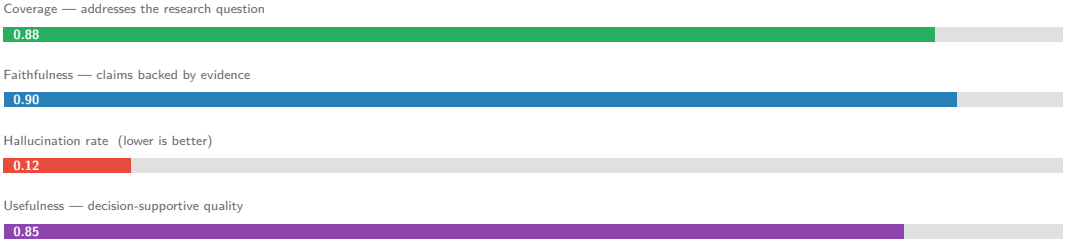
6 themes ▪ 2 contradictions

Performance ▪ Data efficiency ▪ Interpretability ▪ Hybrid models

Report

~2 100 words ▪ 7 sections

Inline citations ▪ Mermaid diagram ▪ PDF export



What we built

- 4-agent pipeline: Search → Synthesis → Report → Evaluate
- FastAPI + SSE streaming + SQLite persistence
- Next.js web UI, REST API, Telegram bot
- Built-in 4-criterion quality rubric
- Groq primary + HuggingFace fallback

From one question to a scored, cited, reproducible report — automatically.

Thank you

Questions?

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